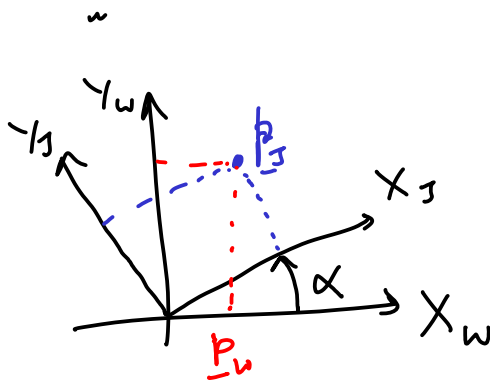


Rotation matrix



Jetbot w.r.t.
World is α

$$\underline{p}_w = {}^w R_J \underline{p}_J$$

$$\begin{bmatrix} x_w \\ y_w \end{bmatrix} = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} x_J \\ y_J \end{bmatrix}$$

① Orthonormal matrix

$${}^w R_J(\alpha)$$

$${}^w R_J^{-1} = {}^w R_J(-\alpha)$$

$${}^w R_J^T {}^w R_J = \underline{I}$$

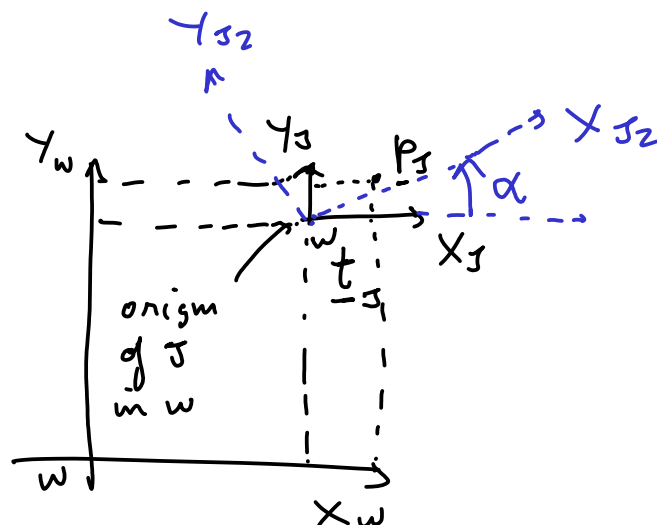
$${}^w R_J^{-1} = {}^w R_J^T = {}^J R_w$$

$$\textcircled{2} \det({}^w R_J) = 1$$

Rotation + translation

$$\underline{p}_w = \underline{p}_J + {}^w \underline{t}_J$$

$$\underline{p}_w \xleftarrow{{}^w \underline{t}_J} \underline{p}_J \xleftarrow{R(\alpha)} \underline{p}_{J_2}$$



$$\textcircled{1} \underline{p}_J = {}^J R_{J_2}(\alpha) \underline{p}_{J_2}$$

$$\textcircled{2} \underline{p}_w = \underline{p}_J + {}^w \underline{t}_J$$

$$\underline{p}_w = {}^J R_{J_2}(\alpha) \underline{p}_{J_2} + {}^w \underline{t}_{J_2}$$

↑ Affine transformation

Transformation matrix

$$\underline{p}_w = {}^J R_{J_2} \underline{p}_{J_2} + {}^w \underline{t}_{J_2}$$

$$\begin{bmatrix} x_w \\ y_w \end{bmatrix} = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} x_{J_2} \\ y_{J_2} \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_w \\ y_w \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_{J_2} \\ y_{J_2} \\ 1 \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_w \\ y_w \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} \cos \alpha & -\sin \alpha & t_x \\ \sin \alpha & \cos \alpha & t_y \\ 0 & 0 & 1 \end{bmatrix}}_{{}^w T_{J_2}} \begin{bmatrix} x_{J_2} \\ y_{J_2} \\ 1 \end{bmatrix}$$

homogeneous coordinates

${}^w T_{J_2}$

$$\Rightarrow \boxed{\underline{\tilde{p}}_w = {}^w T_{J_2} \underline{\tilde{p}}_{J_2}} \Leftrightarrow \begin{bmatrix} \underline{p}_w \\ 1 \end{bmatrix} = \begin{bmatrix} {}^w R_{J_2} & {}^w \underline{t}_{J_2} \\ \mathbf{0}^T & 1 \end{bmatrix} \begin{bmatrix} \underline{p}_{J_2} \\ 1 \end{bmatrix}$$

Block-wise matrix multiplication works as long as the ^{component} multiplications are valid

Properties of Transformation matrix

① Find the inverse of a transformation matrix using the fact that $R^{-1} = R^T$

$$T = \begin{bmatrix} R_{2 \times 2} & t_{2 \times 1} \\ \underline{0}_{1 \times 2}^T & 1 \end{bmatrix}_{3 \times 3}$$

$$T^{-1} = ?$$

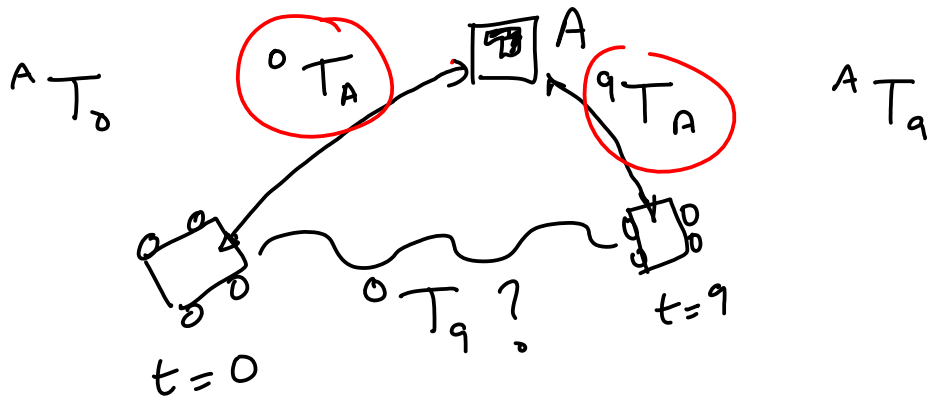
$$\begin{aligned} \underline{y} &= R \underline{x} + \underline{t} \\ \underline{x} &= R^T (\underline{y} - \underline{t}) \end{aligned}$$

$$\begin{aligned} \underline{y} &= T \underline{x} \\ \underline{x} &= T^{-1} \underline{y} \end{aligned}$$

$$\underline{x} = R^T \underline{y} - [R^T \underline{t}]_{2 \times 1}$$

$$\begin{bmatrix} \underline{x} \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} R_{2 \times 2}^T & -R_{2 \times 1}^T \underline{t} \\ \underline{0}_{1 \times 2}^T & 1 \end{bmatrix}}_{T^{-1}} \begin{bmatrix} \underline{y} \\ 1 \end{bmatrix}$$

$$T = \begin{bmatrix} R & \underline{t} \\ \underline{0}^T & 1 \end{bmatrix} \Rightarrow T^{-1} = \begin{bmatrix} R^T & -R^T \underline{t} \\ \underline{0}^T & 1 \end{bmatrix}$$



$${}^0T_9 = ?$$

$$\underline{x}_0 = {}^0T_9 \underline{x}_9$$

$$\underline{x}_A = {}^A T_9 \underline{x}_9$$

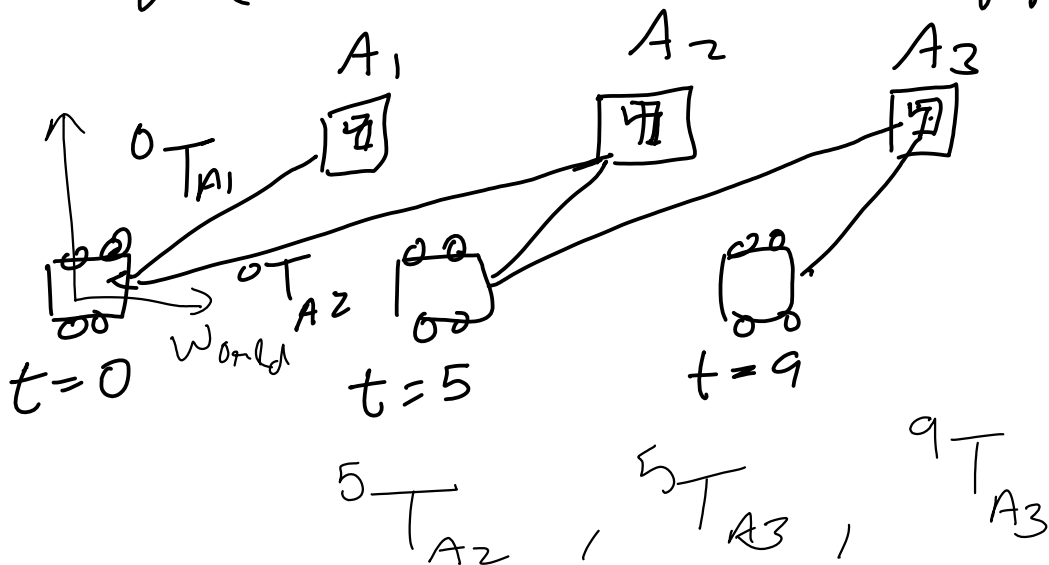
$$\underline{x}_0 = {}^0T_A \underline{x}_A$$

$$\underline{x}_0 = \underbrace{{}^0T_A \quad {}^A T_9}_{{}^0T_9} \underline{x}_9$$

$$\Rightarrow \boxed{{}^0T_9 = {}^0T_A \quad {}^A T_9}$$

Composition of transformations

Mapping (Landmark based mapping)



$${}^0T_{A_3} = ?$$

$${}^0T_9 = ?$$

$${}^5T_{A_2}, {}^5T_{A_3}, {}^9T_{A_3}$$

3D Rotations

$$\begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix} = \underbrace{\begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}}_R \begin{bmatrix} x_s \\ y_s \\ z_s \end{bmatrix}$$

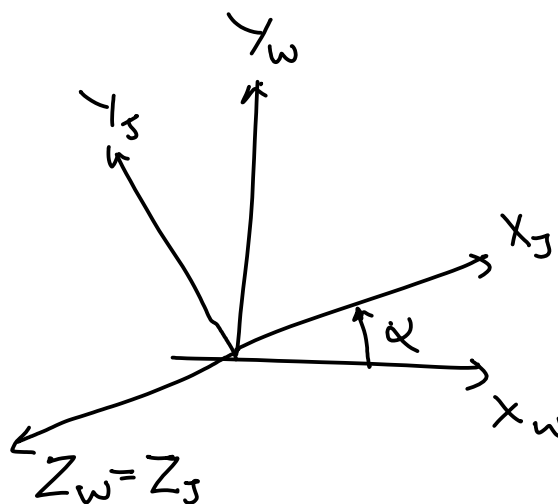
2D Rot

① Orthonormal $R^T R = I$

$$R^{-1} = R^T$$

columns are unit vectors that are $\perp$ to each other

② $\det(R) = 1$



We can rotate in 2D and compose 2D rotations to form 3D rotations.

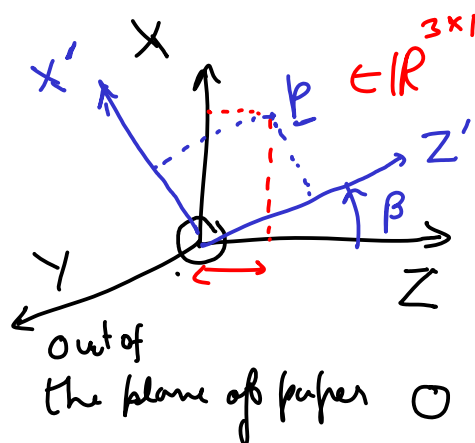
$$\rightarrow R_z(\alpha) = \begin{bmatrix} R_{2D}(\alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3 \times 3} = R_{X-Y \text{ plane}}(\alpha)$$

↖ around z-axis

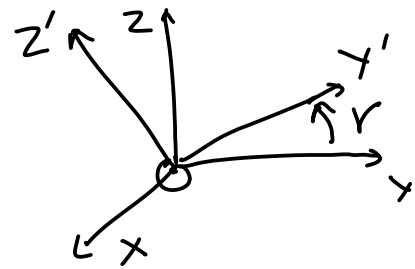
$$R_y(\beta) = \begin{bmatrix} \cos \beta & 0 & \sin \beta \\ 0 & 1 & 0 \\ -\sin \beta & 0 & \cos \beta \end{bmatrix}$$

$$\begin{matrix} -1 & & \\ X & Y & \rightarrow Z \\ & & Z \times X \rightarrow Y \\ & & -1 \end{matrix}$$

$$\underset{3 \times 1}{p'} = \underbrace{R_y(\beta)}_{3 \times 3} \underset{3 \times 1}{p}$$



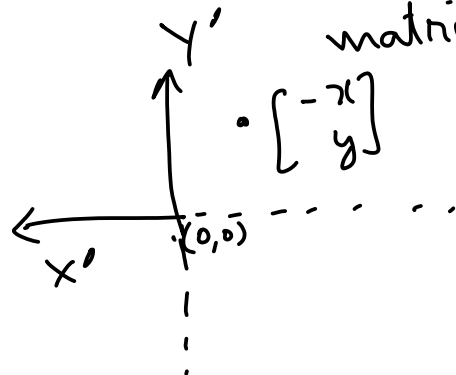
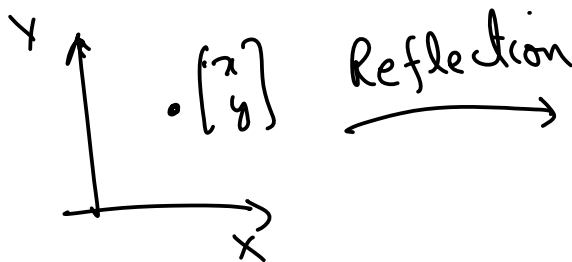
$$R_x(r) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos r & -\sin r \\ 0 & \sin r & \cos r \end{bmatrix}$$



In $n-D$ any matrix that is Orthogonal and has determinant = 1 is a valid Rotation matrix

$$U^T U = I \Rightarrow \det(U) = +1 \text{ or } -1$$

↑
Reflection matrices



$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

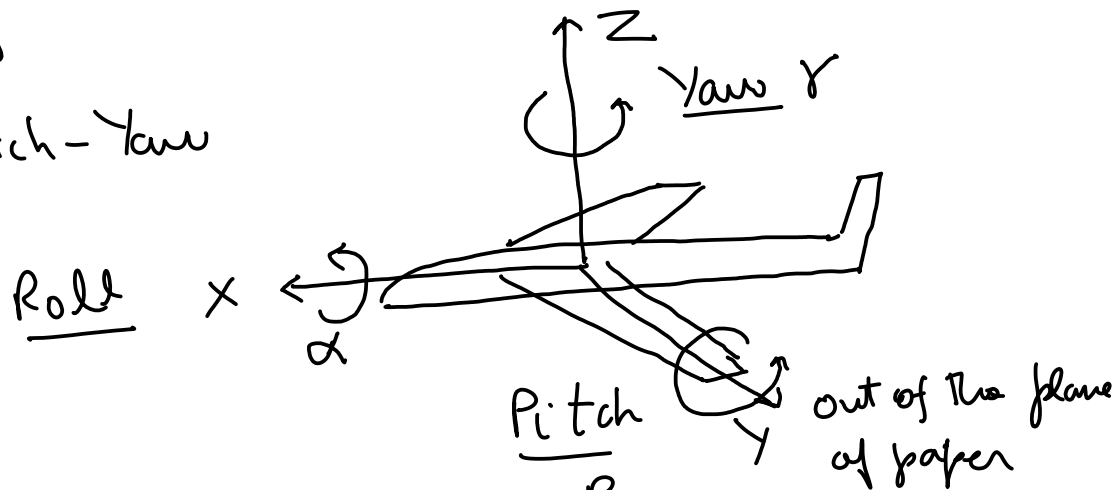
Reflection $\neq$ Rotation

Orthogonal

$$\det(\) = -1$$

Aeronautics

Roll - pitch - Yaw



$$R(\alpha, \beta, \gamma) = R_x(\alpha) R_y(\beta) R_z(\gamma)$$

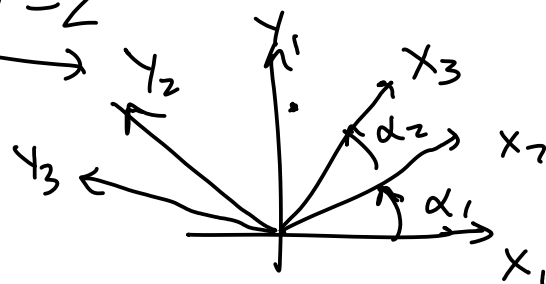
Euler angles

(12 possible

Euler angles)

Axis aligned
Rotations

X-Y-Z



$$\underline{p}_2 = {}^2R_3 \underline{p}_3$$

$$\underline{p}_1 = {}^1R_2 \underline{p}_2$$

$$\underline{p}_1 = {}^1R_2(\alpha_1) {}^2R_3(\alpha_2)$$

roll-pitch-yaw
 Euler angles : X-Y-Z, Y-Z-X, ...
 $3! = 6$
 : X-Y-X, X-Z-X, Y-X-Y, Y-Z-Y
 ...
 6 way

12 ways

Degrees of Freedom in 3D Rotation matrix

(1) Orthonormality

(2) $\det(R) = +1$

Dof
 = count the scalars needed to represent a thing
 - number of constraints

How many scalars in 3D Rotation matrix = ? = 9 scalars

(2) $R^T R = I \leftarrow$ how many constraints? $A_{3 \times 3} \stackrel{?}{=} 3 \times 1 \stackrel{?}{=} 1_{3 \times 1}$

$$\begin{bmatrix} \underline{r}_1^T \\ \underline{r}_2^T \\ \underline{r}_3^T \end{bmatrix} \begin{bmatrix} \underline{r}_1 & \underline{r}_2 & \underline{r}_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \underline{r}_1^T \underline{r}_1 & \underline{r}_1^T \underline{r}_2 & \underline{r}_1^T \underline{r}_3 \\ \underline{r}_2^T \underline{r}_1 & \underline{r}_2^T \underline{r}_2 & \underline{r}_2^T \underline{r}_3 \\ \underline{r}_3^T \underline{r}_1 & \underline{r}_3^T \underline{r}_2 & \underline{r}_3^T \underline{r}_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

6 constraints

all cols are $\perp$

all cols are unit vectors

$$\text{DOF} = 9 \text{ scalars} \begin{matrix} \uparrow \\ \text{free variables} \end{matrix} - 6 \text{ constraints} = 3 \text{ DOF}$$

$$(2) \det(R) = +1$$

$$\det(U) \in \{+1, -1\}$$

$\uparrow$
orthonormal

Euler angle $X-Y-Z$ $\longleftrightarrow$ Rotation matrix?
 α, β, γ $\longleftrightarrow$ R

$$R = R_z(\gamma) R_y(\beta) R_x(\alpha)$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_\alpha & -s_\alpha \\ 0 & s_\alpha & c_\alpha \end{bmatrix} \begin{bmatrix} c_\beta & 0 & s_\beta \\ 0 & 1 & 0 \\ -s_\beta & 0 & c_\beta \end{bmatrix} \begin{bmatrix} c_\gamma & -s_\gamma & 0 \\ s_\gamma & c_\gamma & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$R_x(\alpha) \qquad R_y(\beta) \qquad R_z(\gamma)$

$$c_\alpha = \cos \alpha$$

$$s_\alpha = \sin \alpha$$

$$R = \begin{bmatrix} R_x(\alpha) & & \\ & R_y(\beta) & \\ & & R_z(\gamma) \end{bmatrix} = \begin{bmatrix} c_\beta c_\gamma & -s_\gamma c_\beta & s_\beta \\ -s_\gamma & c_\gamma & 0 \\ -s_\beta s_\gamma & +s_\beta c_\gamma & c_\beta \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}}_R = \begin{bmatrix} C_\beta C_\gamma & -S_\gamma C_\beta & S_\beta \\ C_\alpha C_\gamma - S_\alpha S_\beta C_\gamma & C_\alpha C_\gamma - S_\alpha S_\beta S_\gamma & -S_\alpha C_\beta S_\gamma \\ S_\alpha S_\gamma - C_\alpha S_\beta C_\gamma & S_\alpha C_\gamma + C_\alpha S_\beta S_\gamma & C_\alpha C_\beta \end{bmatrix}$$

$$\frac{-r_{12}}{r_{11}} = \frac{S_\gamma C_\beta}{C_\gamma C_\beta} = \tan \gamma$$

$$\Rightarrow \gamma = \arctan 2(-r_{12}, r_{11}) \quad \left| \begin{array}{l} \text{assuming} \\ C_\beta > 0 \end{array} \right.$$

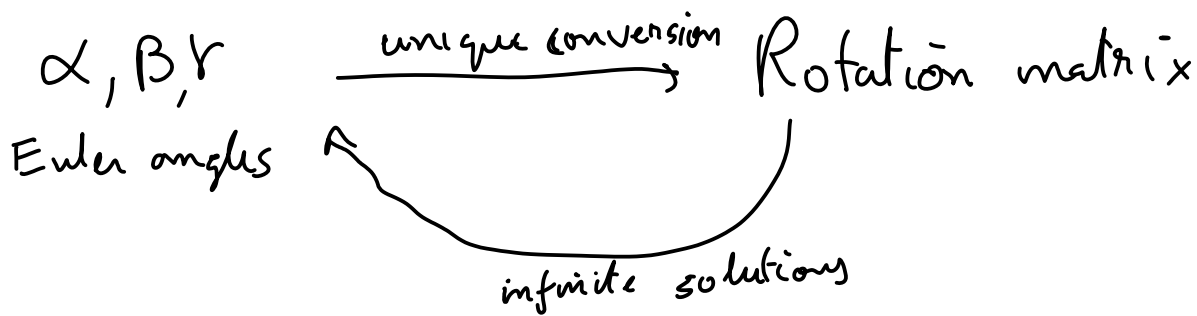
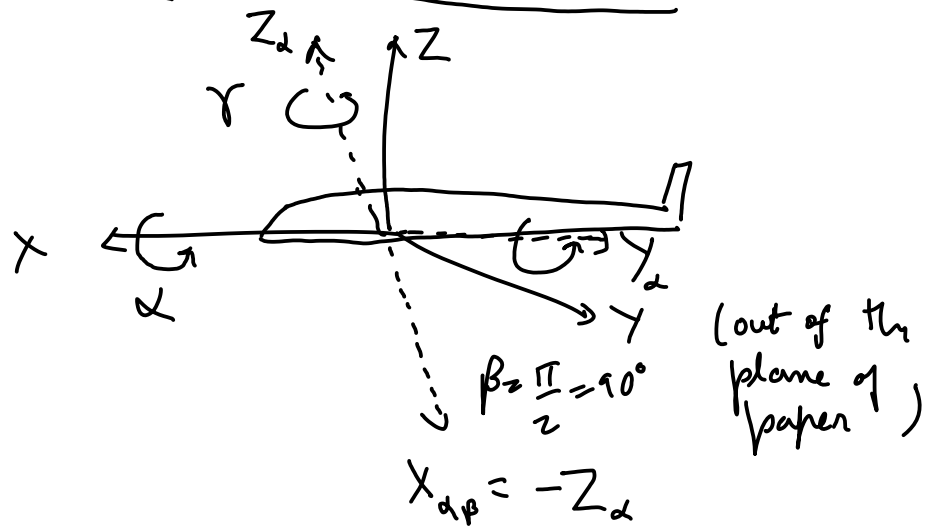
$$\Rightarrow \alpha = \arctan 2(-r_{23}, r_{33}) \quad \left| \begin{array}{l} \text{assuming} \\ C_\beta > 0 \end{array} \right.$$

$$C_\beta = \sqrt{r_{23}^2 + r_{33}^2}$$

$$\begin{aligned} \beta &= \arctan 2(r_{13}, C_\beta) \\ &= \sin^{-1}(r_{13}) \end{aligned}$$

Rot matrix $\rightarrow$ Euler angles

Problem with Euler angles : Gimbal lock



For example

$\alpha =$

$$30^\circ, \beta = 90^\circ, \gamma = 61^\circ$$

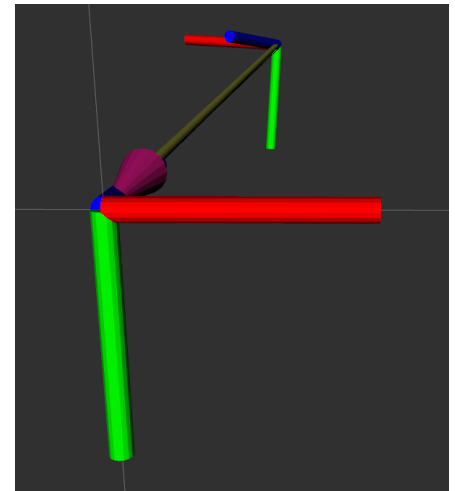
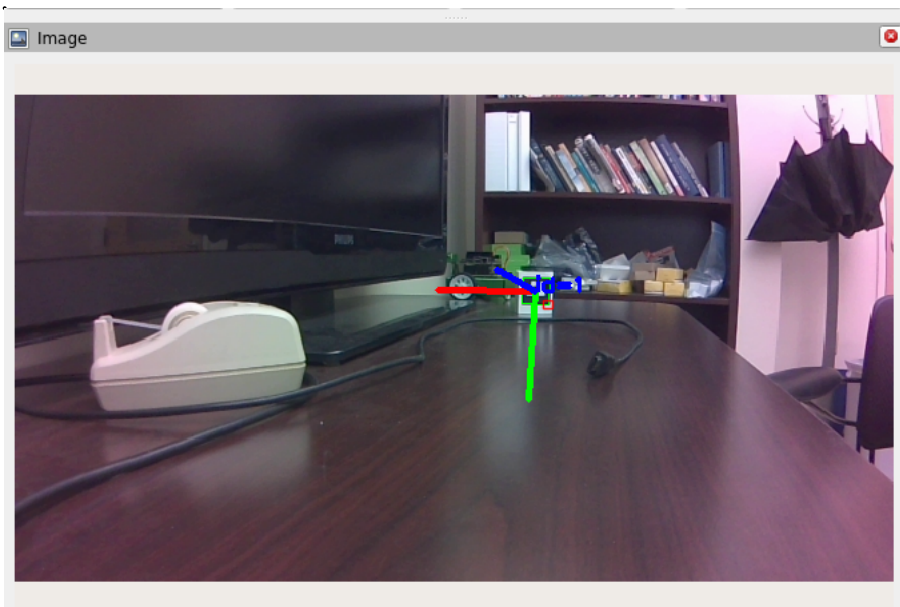
Rotation matrix

$$\alpha = 60^\circ, \beta = 90^\circ, \gamma = 31^\circ$$

$$\text{all } \alpha + \gamma = \boxed{91^\circ}$$

will lead to the

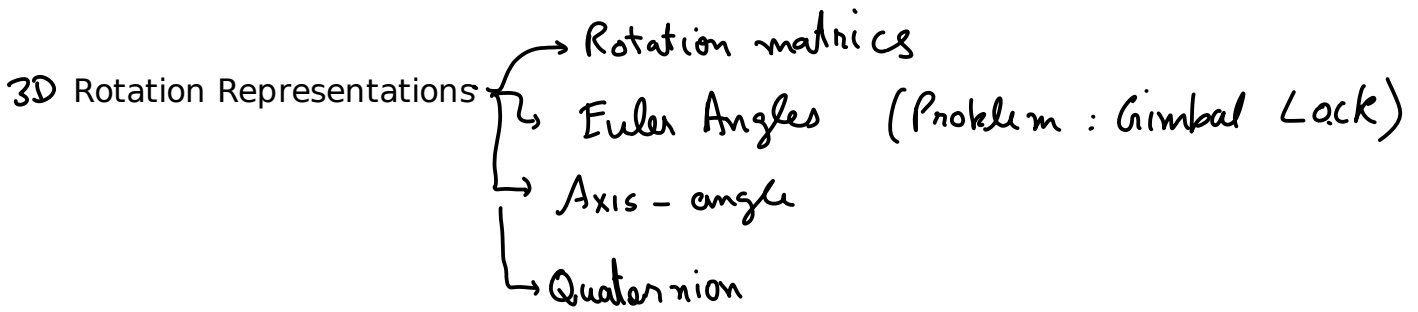
same rotation matrix when $\boxed{\beta = 90^\circ}$



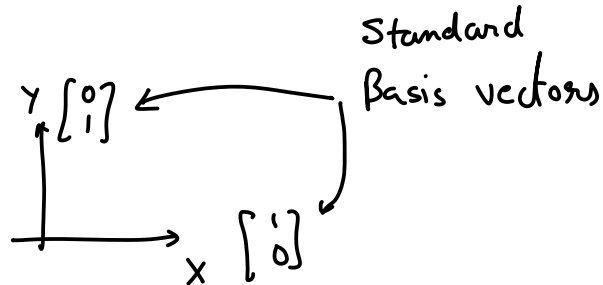
```

root@nano-4gb-jp45:/home/jetbot/ece417# ros2 topic info /
aruco_detections
Type: aruco_opencv_msgs/msg/ArucoDetection
Publisher count: 1
Subscription count: 0

root@nano-4gb-jp45:/home/jetbot/ece417# ros2 topic echo --once /
aruco_detections
header:
  stamp:
    sec: 1727998810
    nanosec: 924374790
  frame_id: /v4l_frame
markers:
- marker_id: 1
  pose:
    position:
      x: 0.08918172498053901
      y: -0.10849999597426438
      z: 0.980432215194246
    orientation:
      x: -0.02973468393320008
      y: 0.9811997541144667
      z: -0.03227342049856027
      w: -0.1879396643245553
boards: []
---
```

Basis vectorsBasis vectors

e.g. Coordinate axis



→ For a given vector space basis vectors are a set of vectors that "span" the vector space

vector space = $\mathbb{R}^2$ = space of all 2D Real vectors

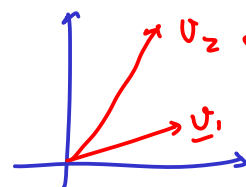
$\underline{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\underline{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $\{\underline{x}, \underline{y}\}$ "spans" $\mathbb{R}^2$

"spans" is the set of all vectors you can obtain by a linear combination of a set of vectors

$\{\underline{x}, \underline{y}\}$ spans $\mathbb{R}^2$ $\mathbb{R}^2 = \{ \underbrace{\alpha \underline{x} + \beta \underline{y}}_{\text{Linear combination}} : \alpha, \beta \in \mathbb{R} \}$

Orthonormal Basis Vectors

- ① Basis vectors that are mutually $\perp$ (orthogonal/perpendicular)
- ② and are unit vectors

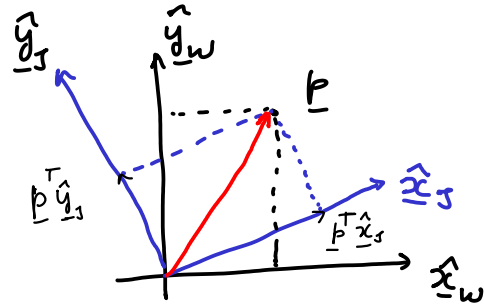


Any two vectors can form a set of Basis vectors as long as they are not parallel.

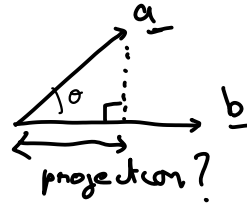
Relationship b/w Orthonormal Basis vector and Rotation matrices

$\hat{x}$ hat is used to denote unit vectors

Projection of a vector?



$$\text{proj}(\underline{a}, \underline{b}) = \frac{\underline{a}^T \underline{b}}{\|\underline{b}\|} \leftarrow \begin{array}{l} \text{dot product} \\ \text{magnitude} \end{array} = \|a\| \cos \theta$$



$$\underline{p} = (\underline{p}^T \hat{x}_s) \hat{x}_s + (\underline{p}^T \hat{y}_s) \hat{y}_s$$

$$\Rightarrow \underline{p}_{2 \times 1} = \underbrace{\begin{bmatrix} \hat{x}_s & \hat{y}_s \end{bmatrix}}_{\text{Basis matrix } B_s} \underbrace{\begin{bmatrix} \underline{p}^T \hat{x}_s \\ \underline{p}^T \hat{y}_s \end{bmatrix}}_{\text{coordinates } \underline{p}_s}$$

$$\Rightarrow \underline{p}_{2 \times 1} = B_s \underline{p}_s$$

for any set of Basis vectors

$$\underline{p} = B_w \underline{p}_w \quad \text{where } B_w = \begin{bmatrix} \hat{x}_w & \hat{y}_w \end{bmatrix} \quad \underline{p}_w = \text{coordinates of } \underline{p} \text{ in } B_w$$

$$\Rightarrow \underline{p} = B_s \underline{p}_s = B_w \underline{p}_w$$

To find $\underline{p}_w$ in terms of $\underline{p}_s$ use orthonormality of B_w

$$\underline{p}_w = \underbrace{B_w^T B_s}_w R_s \underline{p}_s$$

Memorize

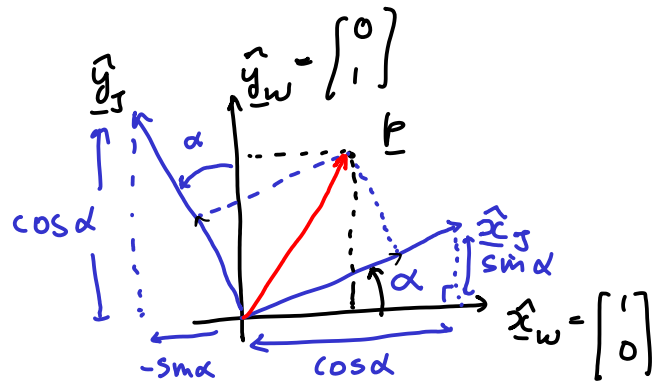
$$B_w = [\hat{x}_w \quad \hat{y}_w] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_{2 \times 2}$$

$${}^w R_J = ? = B_w^T B_J$$

$$\hat{x}_J = \begin{bmatrix} \cos \alpha \\ \sin \alpha \end{bmatrix}, \quad \hat{y}_J = \begin{bmatrix} -\sin \alpha \\ \cos \alpha \end{bmatrix}$$

$$B_J = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} = [\hat{x}_J \quad \hat{y}_J]$$

$${}^w R_J = B_w^T B_J = I^T B_J = B_J = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$$



${}^w R_J = B_w^T B_J$ extends to 3D as well

${}^w B_J$ is 3×3 , B_w, B_J are also 3×3

Example HW or exam problem

Basis vectors

$$\text{Set (1)} \quad (\hat{u}, \hat{v}, \hat{w}) = \underline{\hat{u}} = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}, \quad \underline{\hat{v}} = \begin{bmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}, \quad \underline{\hat{w}} = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$$

$$\text{Set (2)} \quad (\hat{x}, \hat{y}, \hat{z}) = \underline{\hat{x}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \underline{\hat{y}} = \begin{bmatrix} 0 \\ 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}, \quad \underline{\hat{z}} = \begin{bmatrix} 0 \\ 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

Find Rotation matrix that converts coordinates from (1) $\rightarrow$ (2)

and then from (2) $\rightarrow$ (1)

$$^2 R_1 = B_2^T B_1$$

$$= \begin{bmatrix} \underline{\hat{u}} & \underline{\hat{v}} & \underline{\hat{w}} \end{bmatrix}^T \begin{bmatrix} \underline{\hat{x}} & \underline{\hat{y}} & \underline{\hat{z}} \end{bmatrix}$$

$$= \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ -1/2 & -1/2 & -1/\sqrt{2} \\ +1/2 & +1/2 & -1/\sqrt{2} \end{bmatrix}$$